

Algo-Trading Hackathon × Internship Drive

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Designing Risk Intelligence, Not Just Models

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Introduction

Understanding what are we actually looking at and going to do .

FINNIFTY Algorithmic Trading System

Structure-Based Trading with Machine Learning

- 1.Objective:** Develop a profitable algorithmic trading system for FINNIFTYindex
- 2.Approach:** Combine technical structure analysis with XGBoost machine learning
- 3.Period:** August 2021 - April 2024 (404,367 1-minute bars)
- 4.Tools:** Python, XGBoost, Pandas, Scikit-learn

Strategy Overview & Core Intuition

The Core Idea

Strategic Philosophy

Our strategy is built on the principle that price respects structure. Markets move in waves, and optimal entries occur when:

1. Price retraces to key Fibonacci levels (OTE - Optimal Trade Entry)
2. Institutional gaps (Fair Value Gaps) remain unfilled
3. Market regime shows clear directional bias

Why This Works

- Fibonacci 0.5-0.7 Zone: This is where smart money accumulates positions
- Fair Value Gaps (FVG): Imbalances created by rapid price movements that act as magnets
- Market Regime: Trending markets have momentum; we avoid choppy, ranging conditions

Technical Indicators & Features Used

The small elements which help in long run

A. Structure-Based Features

1. Fibonacci Retracement Levels

- Calculated over 100-bar lookback window
- Key levels: 0.382, 0.500, 0.618, 0.700
- OTE Zone: 0.500 - 0.700 (expanded from traditional 0.618-0.786)

2. Fair Value Gaps (FVG)

- Bullish FVG: Gap between candle 1 high and candle 3 low
- Bearish FVG: Gap between candle 1 low and candle 3 high
- Minimum gap: 0.05% of price (relaxed for more signals)
- Persistence: 20 bars forward fill

3. Market Regime Classification

- Uses ADX (Average Directional Index) with 14-period
- Threshold: $ADX > 20$ (lowered from 25 for more signals)
- Classifications: RANGING, TRENDING, STRONG_UPTREND, STRONG_DOWNTREND

Technical Indicators & Features Used

The small elements which help in long run

B. Technical Indicators

- Momentum: RSI, MACD, ROC (5/10/20 periods)
- Trend: SMA & EMA (5, 10, 20, 50, 200 periods)
- Volatility: Bollinger Bands, ATR
- Volume: Volume ratio v 20-period MA, volume trend

C. Time-Based Features

- Hour of day (market hours: 9:15 AM - 3:30 PM)
- Day of week
- Days to month end (expiry awareness)
- Session period (opening, mid-session, closing)

Entry Logic & Signal Generation

Strategy with which we take entry in our trades

Entry Criteria (Confluence Approach)

Step 1: ML Signal

- XGBoost predicts future direction (5-bar horizon)
- Probability threshold: 0.3385 (optmz on validation set)
- Model trained on 253,539 bars (70% of data)

Step 2: Additional Filters

- NOT overstretched (RSI not extreme)
- Volume confirmation (volume surge OR trend)

Step 3: Confluence Filters

Entry requires ANY of the following combinations:

- **Option A: OTE + FVG**

1. Price in OTE zone (0.5-0.7 Fibonacci)
2. Price in Fair Value Gap

- **Option B: Strong Trend**

1. Market regime = STRONG_UPTREND
2. OR Bullish FVG present

- **Option C: Minimum Momentum**

1. ADX > 15 (directional movement)

Exit Logic & Risk Management

Understanding what are we actually looking at and going to do .

Exit Triggers

1. Take Profit Target

- Risk-Reward Ratio: 1:2
- Calculation:

$\text{Risk} = \text{Entry Price} - \text{Stop Loss}$

$\text{Take Profit} = \text{Entry Price} + (2 \times \text{Risk})$

- Example: Entry = 20,000, SL = 19,800
 - Risk = 200 points
 - TP = 20,400 (200 points $\times$ 2)

2. Chandelier Trailing Stop

- Formula: Highest High (22 bars) - (3 $\times$ ATR)
- Trails upward only (locks in profits)
- Adapts to volatility automatically

3. Opposite Signal

- ML probability drops below 0.3 (30%)
- Indicates potential reversal

4. Trend Weakening

- ADX falls below threshold
- Loss of directional momentum

5. Position Sizing

- Fixed: 10% of current capital per trade
- Commission: 0.03% per trade
- Slippage: 0.01% per trade
- Initial Capital: \$100,000

Risk Management Framework

Understanding what are we actually looking at and going to do .

A. Portfolio-Level Risk

- Max Drawdown Tolerance: 10%
- Max Concurrent Positions: 1

B. Execution Risk

- Slippage: 0.01% factored into all trades
- Commission: 0.03% per trade
- Entry/Exit Timing: Market orders with realistic fills

C. Strategy Risk Controls

- Overstretched Filter: Avoid entering in extreme conditions ($RSI > 80$ or < 20)
- Regime Filter: Only trade in trending markets (avoid chop)
- Volume Filter: Ensure liquidity and conviction
- Chandelier Stop: Automatic risk adjustment based on volatility

Machine Learning Architecture

Designing for Signal, Stability, and Scale.

Model Selection: XGBoost Classifier

Why XGBoost?

- Handles non-linear relationships in market data
- Built-in regularization prevents overfitting
- Feature importance for interpretability
- Fast training and prediction
- Robust to imbalanced classes

Feature Engineering Pipeline

1. Raw Data (OHLCV) → 8 columns
2. Structure Features (Fib, FVG, Regime) → +25 features
3. Technical Indicators (SMA, RSI, MACD) → +30 features
4. Time Features (Hour, Day, Expiry) → +8 features
5. Lag Features (Returns, Volume) → +9 features
6. Total: 70+ features → Model ready

Training Process & Data Splits

Understanding what are we actually looking at and going to do .

Chronological Splitting Strategy

- Method: Strict time-series split (no random shuffling) to simulate real-world forecasting.
- Train Set (2021 – Jan 2026)
- Validation Set (Jan 2023 – July 2023):
- Test Set (July 2023 – April 2024)

Data Preparation Pipeline

- Cleaning: Rows with NaN targets or >50% missing features were dropped to ensure data integrity.
- Scaling: Applied StandardScaler to normalize all input features (Mean=0, Variance=1), ensuring indicators like Volume don't dominate oscillating indicators like RSI.
- Imbalance Handling: The dataset was naturally balanced (~50/50 Up/Down split).

Split	Samples	Purpose
Training	256,715	Model learning
Validation	36,673	Perf Vald'n
Test	73,347	Final evaluation

Validation Strategy

“Ensuring Fair and Representative Model Evaluation.

Original Dataset

Total samples: 366,737

Why Hold-Out Validation?

- Training performance alone is misleading
- Models may memorize training patterns (overfitting)
- Validation data simulates unseen future customers

Random Split Risk

- Training data may have irregular split
- Validation and Testing can give different results

Safety Check

- The pipeline explicitly drops rows with NaN targets to ensure data integrity.

Split	Training Size	Validation Size	Testing Size
90 / 5 / 5	330,065	18,336	18,336
70 / 10 / 20	256,715	36,673	73,347

Testing Strategy

Testing That Reflects Real-World Performance.

Evaluate performance using both statistical metrics and financial reality.

Evaluation Metrics (ML Layer)

- ROC-AUC: Primary metric to measure the model's ability to distinguish between "Up" and "Down" moves (Result: 0.5636, indicating weak predictive power).
- Precision vs. Recall: Analyzed the trade-off between capturing moves (Recall) and being correct (Precision).
- Confusion Matrix: Monitored False Positives (bad trades) vs. True Positives to understand error distribution.

Backtesting Engine (Financial Layer)

- Simulation: Realistic trade execution using the Test Set (unseen data).
- Constraints: Included transaction costs (Commission: 0.03%, Slippage: 0.01%) to punish over-trading.
- Risk Management: Implemented Chandelier Exits (Trailing Stop) and fixed Risk:Reward (1:2) targets.
- Reality Check: The backtest revealed a -55.30% return, highlighting that while the ML mechanics worked, but the underlying signal was not profitable.

Optuna Hyperparameter Tuning

Searching the Space Where Performance Actually Lives

Optimization Strategy

- 50 Optuna trials per model
- Tuned XGBoost

Hyperparameter Focus

- Model Capacity: max_depth , max_delta_step , gamma
- Learning Dynamics: learning_rate , n_estimators
- Regularization: reg_alpha , reg_lambda , min_child_weight
- Sampling: subsample , colsample_bytree ,
colsample_bylevel

Optuna Hyperparameter Tuning

Searching the Space Where Performance Actually Lives

XGBoost Hyperparameters:

- max_depth: 6
- learning_rate: 0.05
- n_estimators: 1000
- min_child_weight: 3
- subsample: 0.8
- colsample_bytree: 0.8
- colsample_bylevel: 0.5
- gamma: 0.1
- reg_alpha: 0.1
- reg_lambda: 1.0
- max_delta_step: 2

Performance: Value = 0.6291

XGBoost Hyperparameters:

- max_depth: 9
- learning_rate: 0.015816981413469905
- n_estimators: 400
- min_child_weight: 3
- subsample: 0.6389274219025062
- colsample_bytree: 0.9987349142404867
- colsample_bylevel: 0.7705609360826743
- gamma: 0.28155316324336543
- reg_alpha: 1.7346201315788978
- reg_lambda: 3.2706207110759644
- max_delta_step: 4

Performance: Value = 0.6746

Key Findings & Final Results

Made for learning , not just score.

1. Critical Failure

- Net Result: \$45,248 (Total Loss)
- Drawdown: -45.25% loss of total equity.
- Reliability: 0 Profitable Months
- Time-Frame should have been 5-min.
- Option greeks (Theta decay)

2. The Core Problems

- Win Rate: 9.8% (Only 1 in 10 trades makes money).
- Risk/Reward Inversion:
 - Avg Win: \$5.81
 - Avg Loss: \$10.95
 - Result: You lose nearly \$2 for every \$1 you make.
- Over-Trading: 4,867 trades suggests signal noise, not signal quality.

3. The "Exit" Paradox

- Data Conflict: The system claims 58% of exits were "Take Profits," yet the win rate is only 9%.
- Conclusion: The "Take Profit" targets are likely set so tight they fail to cover spread/commissions, turning technical wins into realized losses.

*Thank
You*

Questions?

Project Repository:

<https://github.com/OmChauhan2023/Falcon->

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